DEEPCORE LABS · DIGITAL MINE ON OPENEGIZ 2026

MineGuard X

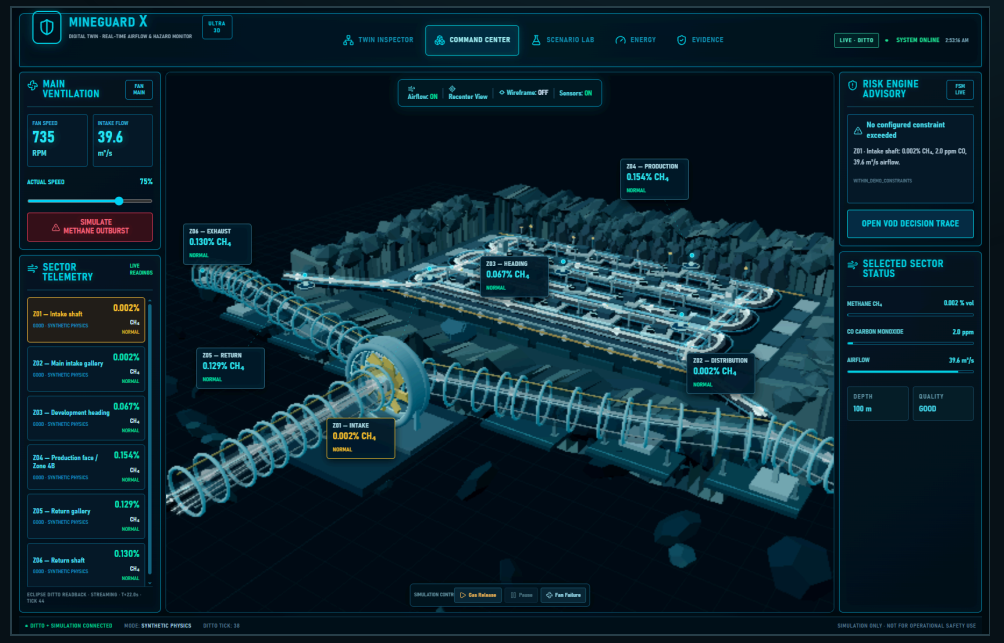
A physics-informed digital twin of an underground coal-mine ventilation section

Six connected ventilation zones and two fans live as Eclipse Ditto twins. A deterministic airflow and gas model publishes through OpenEgiz MQTT, and every value in the interface is read back from Ditto. Operators, ventilation engineers and reliability engineers use it to decide how to ventilate safely and efficiently.

SIMULATION ONLY

FICTIONAL, DOCUMENTED PARAMETERS

OPENEGIZ v1.0.4 + ECLIPSE DITTO



CONCEPT

Ventilation, methane, fan reliability and energy are one problem

Enterprise

A gassy coal-mine section: intake shaft, main intake gallery, development heading, production face, return gallery, return shaft; exhausting main fan and auxiliary heading fan.

Users

Control-room operator (what is happening, where) · ventilation engineer (which fan setting is safe, at what energy) · reliability engineer (what a fan failure does; did the requested response happen).

Decision

When rising CH_4/CO requires stronger ventilation, a production stop or evacuation; which fan speeds satisfy every constraint, and which uses least energy; whether the response was achieved.

Why

Changing one fan changes airflow, and therefore methane dilution, in every connected zone. Methane accidents such as Kostenko (Karaganda, 2023) show the stakes. Main fans are large electrical loads, but safety must always outrank energy.

MINE MODEL

Six zones, six directed airways, two fans — nine Ditto twins

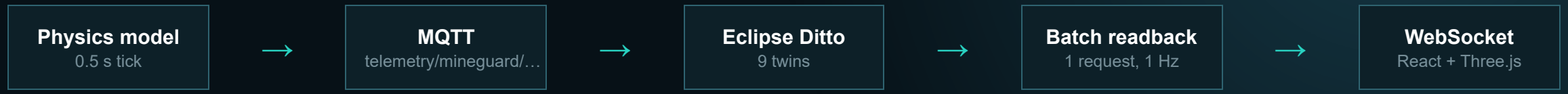
> `Z01 Intake → Z02 Distribution → { Z03 Heading (aux fan) ‖ Z04 Production face } → Z05 Return → Z06 Return shaft (main fan)`

THING	FEATURES PUBLISHED EVERY SECOND
`mineguard:mine-01`	max CH_4, max CO, total airflow, fan power, per-airway flow E01–E06, tick, paused
`mineguard:zone-z01…z06`	methane (% vol), co (ppm), airflow (m^3/s), risk (state, reasons, quality), safety_interlock (trip, evacuation, fan requests, equipment_io=false)
`mineguard:fan-main / fan-aux`	speed and requested speed (% rated), rpm, power (kW), state RUNNING/FAULTED

Topology is validated at startup (connected, acyclic, flow-balanced): `mineguard/data/mine_topology.json`. Twins share `_parents: mineguard:mine-01` and the platform policy `default:basic_policy`.

ARCHITECTURE

The UI reads the platform, not the simulator



Official path preserved

Complete `modify /features` messages to the OpenEgiz broker; Ditto events → Telegraf → InfluxDB → Grafana. The official `make smoke` passes with MineGuard publishing.

Verified readback

A publication counts only when all nine twins return the same run ID and tick. The UI state is rebuilt from that Ditto JSON (`ECLIPSE_DITTO_READBACK`). Scenario commands are guarded and simulation-only.

PHYSICS

Conservative, deterministic, explicit units

$$\text{Airway: } \Delta p = R \cdot Q \cdot |Q| \quad \cdot \quad \text{Fan: } \Delta p = p_0 \cdot s^2 - K \cdot Q^2 \quad \cdot \quad \text{Power: } P = P_{rated} \cdot s^3$$

$$\text{Zone gas balance: } V \cdot dC/dt = G + \Sigma\, Q_{in} \cdot C_{in} - \Sigma\, Q_{out} \cdot C \quad \text{(methane and CO)}$$

Network solve

Bisection on the common branch pressure; flow is conserved at every junction. Main-fan failure → zero network flow.

Stable transport

Explicit sub-steps bounded by CFL 0.5; values outside [0,1] are rejected, never clipped. Live residual ≈ 2e-15 m³.

Units

Methane is a fraction internally and % volume outside (0.8 % = 8,000 ppm); CO ppm; airflow m³/s; power kW; energy kWh.

RISK AND SIMULATED SAFETY

Uncertainty and safety come before optimization

STATE	TRIGGER (DEMONSTRATION)	SIMULATED RESPONSE
ADVISORY	CH_4 ≥ 0.5 % or CO ≥ 25 ppm	ventilation request; optimizer still allowed
WARNING	CH_4 ≥ 1.0 %, CO ≥ 35 ppm (NIOSH REL), or airflow < 5 m^3/s	production stop (trip power), optimizer disabled
EMERGENCY	CH_4 ≥ 1.5 %, CO ≥ 50 ppm (OSHA PEL), or main fan failed	evacuation advisory; healthy fans driven to 100 %; failed fan excluded
UNKNOWN_DATA	stale or invalid telemetry	treated as unsafe: production stop and evacuation, never a safe reading

The 1.0 % and 1.5 % tiers mirror US 30 CFR 75.323(b) action levels, for demonstration only, not as a compliance claim. Downgrades use hysteresis; every transition carries reason codes.

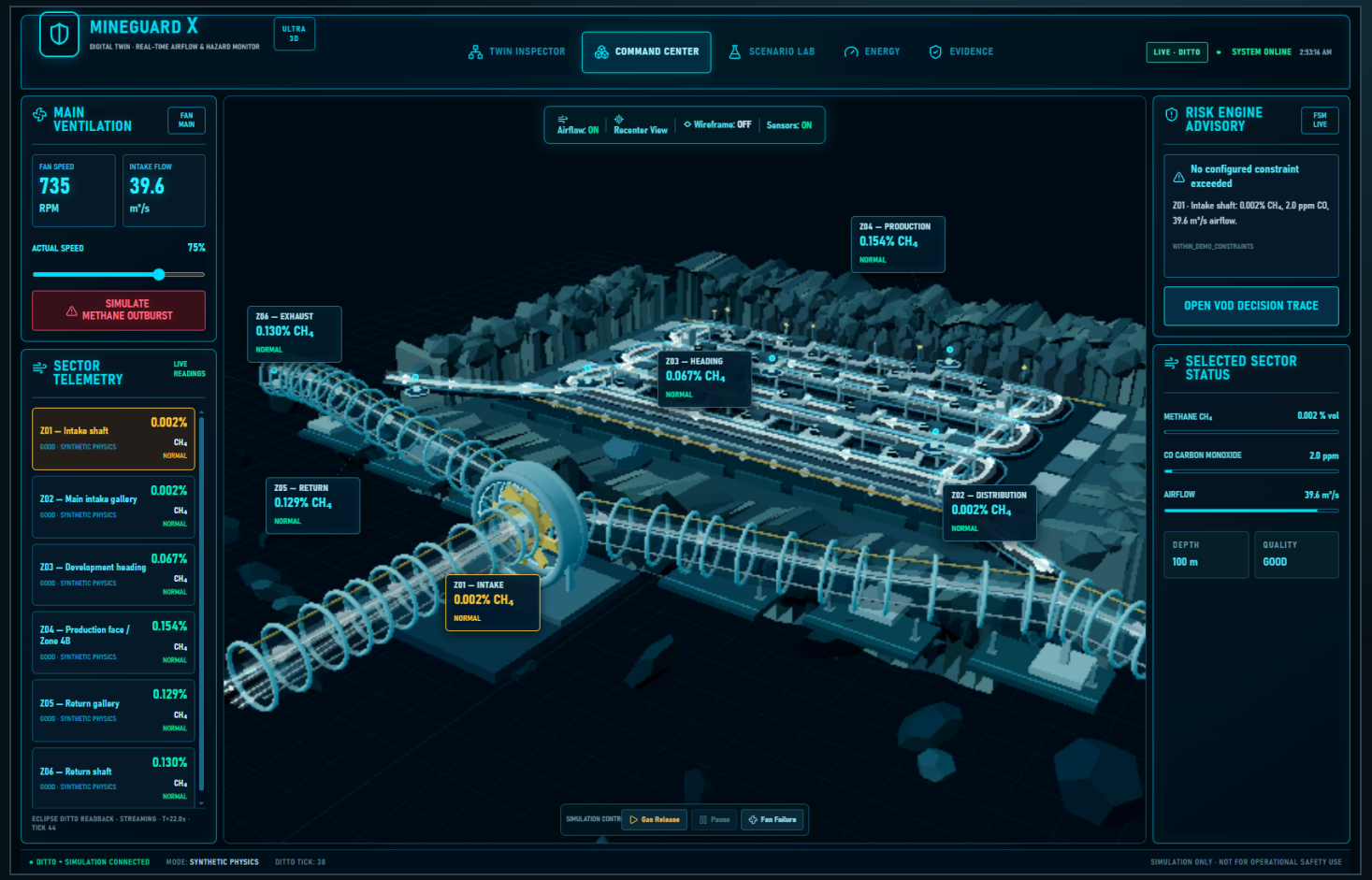
MINE VISUALIZATION

Command Center: 3D cutaway driven by Ditto state

Zone markers take the colour of the twin's risk state. Labels show live CH_4 and the risk state. A false-colour methane volume grows with CH_4. Airflow particles move with the measured intake flow and stop at zero. The fan rotor follows actual RPM and its rim turns red on a fault.

The rails show main-fan RPM, intake flow, every zone's CH_4, the risk advisory, and the selected zone's CH_4, CO and airflow.

On CPU-only WebGL a lite profile applies automatically: 88 draw calls instead of 1,229, Lambert shading and no backdrop blurs.



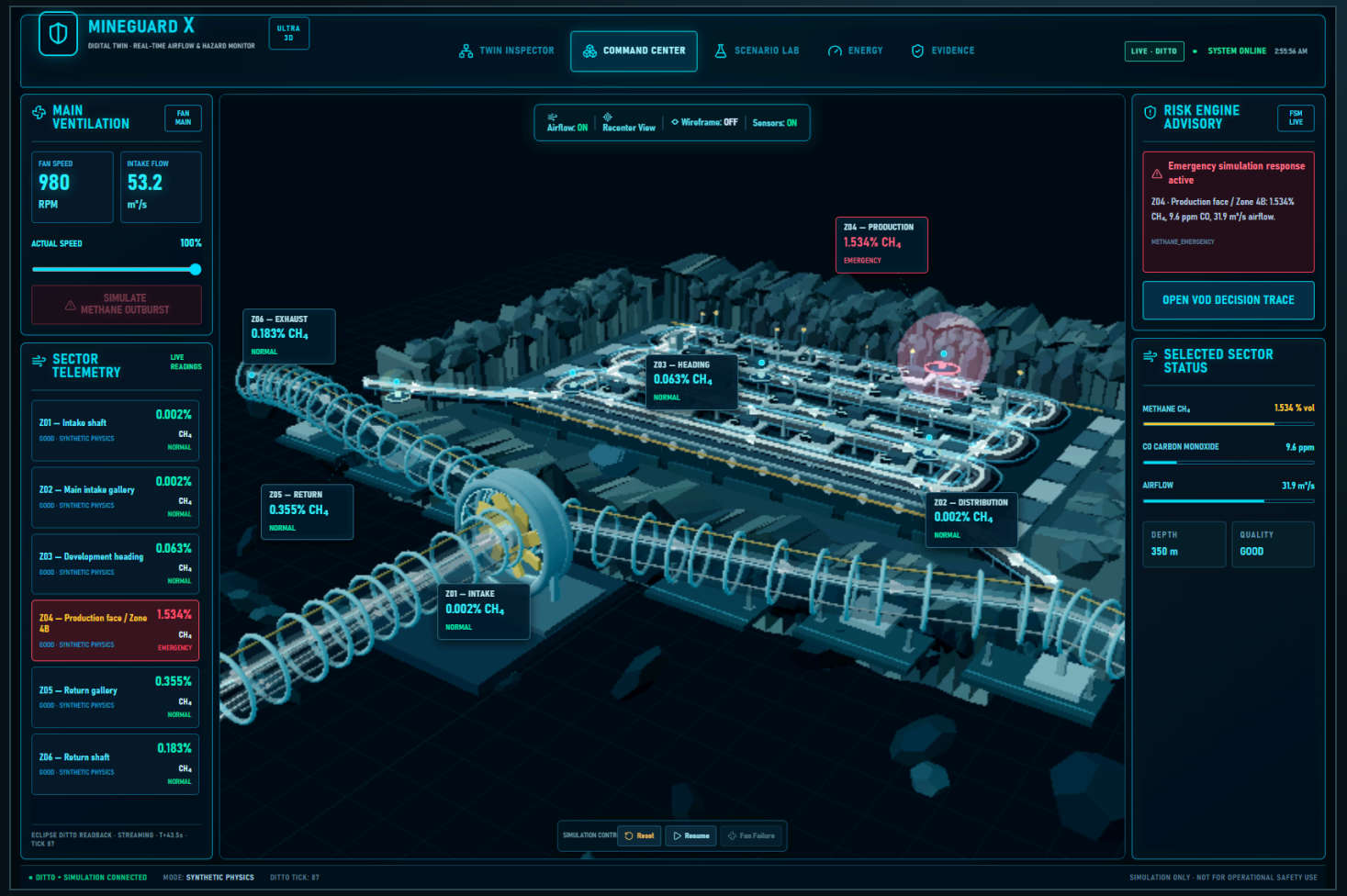
WHAT-IF SCENARIO S1

Production-face methane release

The Z04 source rises from 0.04 to 0.8 m³/s for 180 s. Methane accumulates through the mass balance:

STAGE	TICK	S	Z04 CH_4
ADVISORY	19	9.5	0.53 %
WARNING	48	24.0	1.02 %
EMERGENCY	87	43.5	1.53 %

At EMERGENCY, Ditto shows the Z04 interlock tripped and both fans at 100 % (model feedback ACHIEVED).
Undo: Reset restores the exact tick-0 baseline.



Sampled from the 1 Hz Ditto readback by the scenario evidence runner

WHAT-IF SCENARIO S2

Main ventilation fan failure

At T+10 s the main fan faults. Ditto reads back **0 RPM, FAULTED**; network airflow is 0 m³/s; every zone goes to EMERGENCY with reason `FAN_FAILED`; the failed fan is excluded from responses; methane starts to accumulate; the optimizer reports `NO_FEASIBLE_SAFE_OPTIMIZATION`.

Undo: Reset, or the fault clears at T+90 s. Pause and resume freeze and continue the simulated clock.

Both scenarios run from the Command Center or the Scenario Laboratory without code changes.



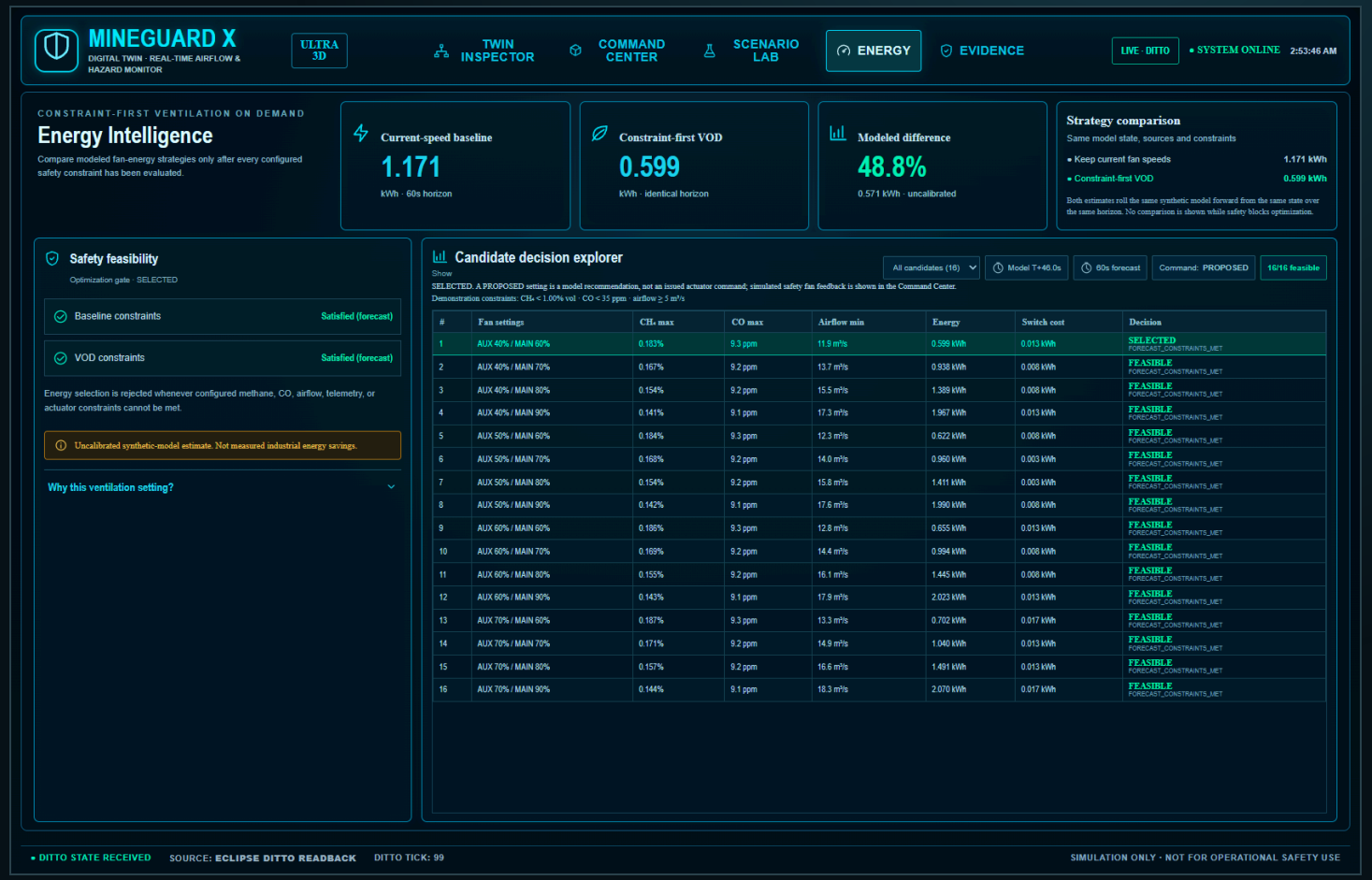
ENERGY MODULE

Constraint-first ventilation on demand

1. Check risk state and data quality
2. Enumerate ramp-limited fan settings (±20 %)
3. Forecast airflow, CH_4 and CO for 60 s
4. Reject every constraint violation
5. Pick the minimum energy plus switching penalty

From the initial state: **1.171** kWh at current speeds vs **0.599** kWh selected (main 60 %, aux 40 %) — **48.8 %**, with all 16 candidates feasible.

Uncalibrated synthetic model estimate, not measured savings. Disabled while any zone is WARNING, EMERGENCY or UNKNOWN_DATA.



EVIDENCE

Claims are tested, and checked again at runtime

Source gates

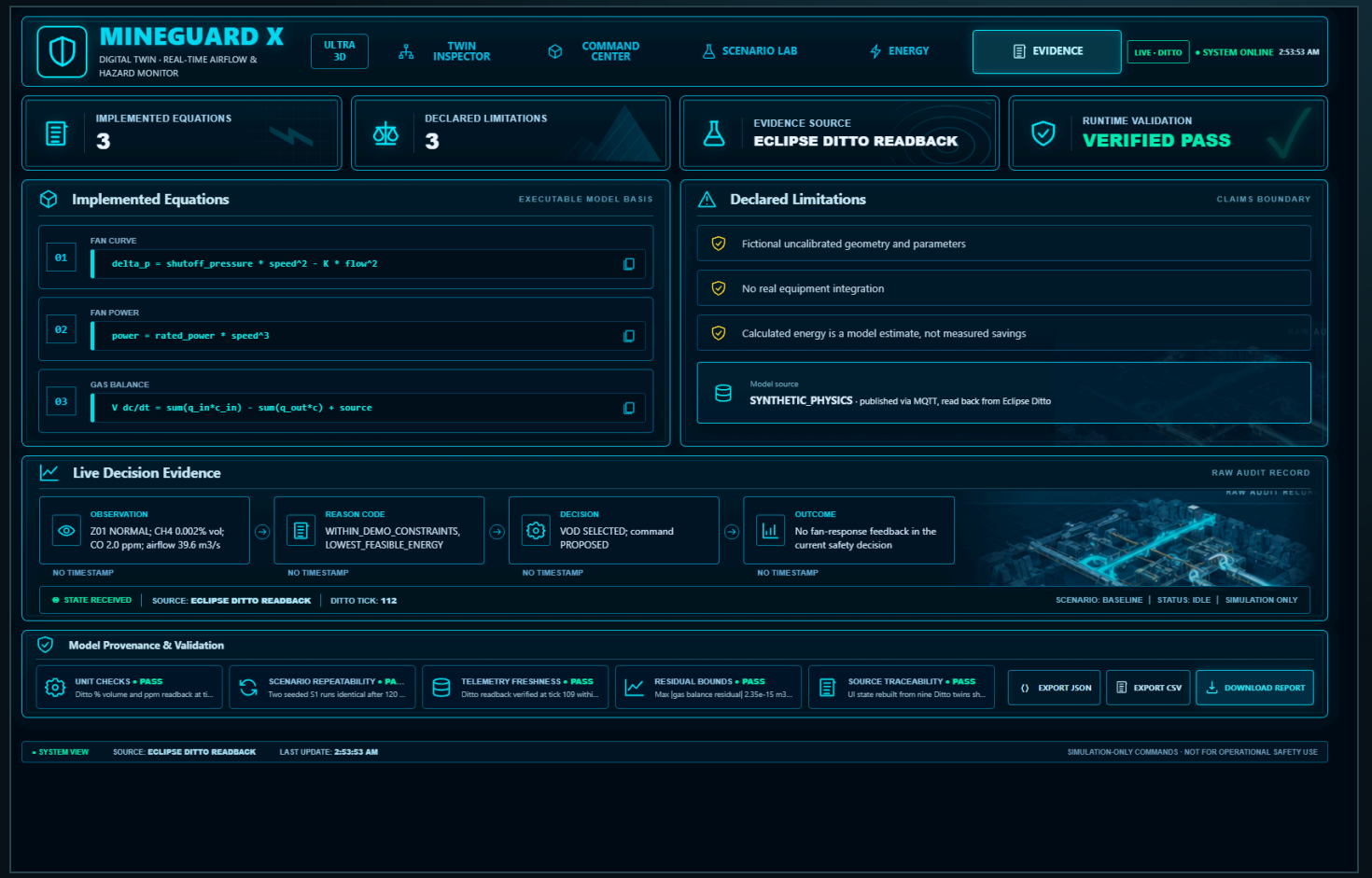
198 backend tests (pytest), Ruff, strict mypy · 31 frontend tests (Vitest), TypeScript, ESLint, production build.

Live gates

official `make smoke` + MineGuard WebSocket and Ditto checks; S1 emergency, interlock and reset; headless-Chrome checks of all five routes on desktop and mobile with no browser errors.

Runtime validation (Evidence page)

Ditto vs model at the same tick: CH_4 error 3.9e-11, CO 3.1e-9 ppm · balance residual 2.2e-15 m^3 · two seeded S1 runs identical, reset exact · readback fresh within 1 s · single-publication traceability.



RUN IT · LIMITS

Three commands on a clean Ubuntu with Docker

```
sudo apt-get update && sudo apt-get install -y make
bash mineguard/scripts/bootstrap.sh   → http://127.0.0.1:8091
```

Limitations

Fictional, uncalibrated mine; steady network flow without leakage or reverse flow; well-mixed zones; quasi-steady fan response; 3D renders at a few frames per second on CPU-only WebGL; no obstruction or sensor-loss scenario in the UI.

Before any real use

Surveyed geometry, calibrated sensors and fans, an OT gateway, authentication, redundancy, alarm management, a hazard analysis and regulatory approval. All actions here are simulated.

FICTIONAL MODEL | SIMULATED ACTIONS | BUILT ON OPENEGIZ